# AI Co-Pilot: Aviation Dataset Pipeline & Feature Engineering Architecture

## STAGE 1: RAW MULTI-MODAL AVIATION DATA INGESTION

**NASA ASRS Database**

6,978 cleaned incident records
Multi-system operational narratives
Expert pilot human factor contexts

**OpenSky Squawk 7700**

832 emergency trajectory segments
High-density ADS-B telemetry
Dynamic 30-min gap boundaries

**Normal Flight Generator**

1,716 nominal baseline trajectories
Climb, cruise, and descent envelopes
Zero-hazard operational profiles

## STAGE 2: DATA ADAPTER PIPELINE

**map_asrs_record()**

Parses flight phase, severity level,
system hazard flags & target SOP action

**map_opensky_record()**

Extracts callsign, aircraft type, ADS-B
spatial track coordinates & kinematic rates

**map_normal_envelope()**

Simulates standard operating envelopes
and baseline aerodynamic limits

**12-Parameter Aerodynamic Feature Synthesizer (Z-Score Standardized on Train Set)**

Kinematic Variables: Altitude (h), Airspeed (Va), Vertical Rate (h_dot), Pitch Angle (θ), Roll Angle (ϕ), Heading (ψ)
Control & Energy: Throttle (N1), Load Factor (G), Runway Distance (d_rw), Wind (Vw), Specific Energy Index (E), Flight Phase Code (κ)
Enforces Zero Lookahead Leakage via Pre-Fitted Train Scaler Parameters

## STAGE 3: UNIFIED SEQUENCE TENSOR

**Leakage-Free Standardized Sequential Tensor: X ∈ ℝ^(9,526 × 15 × 12)**

Total Ingested Sequences: N = 9,526 | Temporal Horizon: T = 15 Timesteps | Feature Channels: F = 12
Temporal Stride: S = 5 timesteps | Uniform Temporal Discretization Across Heterogeneous Trajectories

## STAGE 4: TRAIN / TEST SPLIT (80 : 20)

**Training Partition (80%)**

7,620 Sequential Samples (Fit split only)
Used exclusively for model optimization and scaling statistics

**Held-Out Testing Partition (20%)**

1,906 Sequential Samples (Strictly unseen)
Isolates entire flight trajectories to prevent temporal autocorrelation leakage

## STAGE 5: MULTI-SOURCE CORPUS COMPOSITION & SEVERITY DISTRIBUTION




**Total Ingested Sequences: 9,526 Continuous Windows | 12 Standardized Aerospace Parameters | Zero Data Leakage**